# UberClock Reasearch Conclusion

Resonant-mode thermometry, clock reconstruction and the path toward autonomous compensation

**Project review — 21 October 2026**

## Research objective

UberClock investigates whether several resonant modes of a crystal can provide both a temperature estimate and enough information about the FPGA timebase to synthesize a stable output clock. An external 10 MHz reference is available during development; the final objective is operation based on resonant-mode measurements and a learned oscillator model.

Three modes are tracked: C300 near 10.004 MHz, A100 near 6.270 MHz and C100 near 3.389 MHz. Their different temperature sensitivities are the source of information for separating crystal temperature from a shared clock-scale error.

## Outcome at this checkpoint

| Research component | Conclusion |
|---|---|
| Modal tracking and thermometry | Resonant responses can be located and tracked approximately. Characterization supports temperature estimation. The improved estimator substantially reduces the apparent temperature fluctuation associated with common frequency scaling. |
| Board-clock diagnosis | Independent raw hardware counting confirms a sawtooth in the board/reference frequency ratio: approximately 42 ppm peak-to-peak with a period near 65 s. |
| Autonomous reconstruction | The mode-only clock estimate does not recover this waveform. A stable temperature estimate is insufficient evidence of a correct reconstructed clock. |
| Reference-assisted synthesis | A separate controller measures the reference every 100 ms and updates the DAC2 NCO increment. Processing and data continuity are validated; physical output stability and phase alignment remain unmeasured. |

## Development sequence

The first design used the external reference to estimate the board-clock scale, corrected all three measured modal frequencies, and only then calculated temperature with WLS. The resulting temperature followed the clock's sawtooth pattern. This observation led to explicit separation of common scale and temperature, first through corrected joint WLS, then through raw joint and frequency-ratio estimators.

The expanded derivation preserves that order. It explains why ideal clock correction should work, why an uncanceled common component enters the initial temperature estimate, and what each replacement estimator can and cannot reject. Recorded cases near 19 °C and 23 °C compare all four estimators before the discussion of clock reconstruction, hardware validation and future work.

**Central distinction.** Decoupling the temperature estimate was successful in the retained comparisons. Recovering the physical clock modulation from those same modal measurements was not. The mathematical models and the experimental evidence for these two claims are presented separately.

# 1. Initial WLS and clock-generation concept

## 1.1 The original measurement scale

The first implementation did not estimate temperature independently of the reference. It first estimated the actual FPGA sampling clock from the external 10 MHz input, multiplied every reconstructed modal frequency by the resulting scale factor, and supplied those corrected frequencies to a one-state WLS temperature estimator. The sawtooth observed in that temperature output motivated the subsequent decoupling work.

The initial model accounts for sampling-clock error in frequency measurement and generation. The mode order throughout is C300, A100, C100. Define

$$F_0 = 65\,000\,000 \text{ Hz}, \qquad F = \text{actual UC sampling clock}, \tag{1}$$

$$f_R = 10\,000\,000 \text{ Hz}, \qquad D = 6500, \tag{2}$$

$$F_{\mathrm{DS},0} = F_0/D = 10\,000 \text{ Hz}, \qquad F_{\mathrm{DS}} = F/D. \tag{3}$$

For an analog sinusoid at physical frequency $f$, sampling gives

$$x[n] = \sin\left(2\pi \frac{f}{F} n + \phi\right), \qquad \rho = \frac{f}{F}. \tag{4}$$

An estimator converting normalized frequency to hertz with $F_0$ reports

$$f_{\mathrm{meas}} = F_0 \rho = f \frac{F_0}{F}. \tag{5}$$

Write the fractional clock error and the two reciprocal scale factors as

$$\epsilon = \frac{F - F_0}{F_0}, \qquad C_F = \frac{F}{F_0} = 1 + \epsilon, \qquad s = \frac{F_0}{F} = \frac{1}{1+\epsilon}. \tag{6}$$

Thus

$$f_{\mathrm{meas}} = sf, \qquad f = C_F f_{\mathrm{meas}}, \qquad F > F_0 \implies f_{\mathrm{meas}} < f. \tag{7}$$

For small $\epsilon$, the complete expansion and first-order approximation are

$$\frac{1}{1+\epsilon} = 1 - \epsilon + \epsilon^2 - \epsilon^3 + \cdots, \qquad f_{\mathrm{meas}} \simeq f(1-\epsilon). \tag{8}$$

The error discarded by this approximation is

$$\frac{1}{1+\epsilon} - (1-\epsilon) = \frac{\epsilon^2}{1+\epsilon}, \qquad e_{\mathrm{lin}} = f \frac{\epsilon^2}{1+\epsilon}. \tag{9}$$

At $f = 10$ MHz and $\epsilon = 30 \times 10^{-6}$, $e_{\mathrm{lin}} \simeq 0.009$ Hz. The implemented correction can use the exact ratio instead of this approximation.

**Sign convention.** $C_F$ corrects a nominal-scale measurement into physical hertz; $s$ is the inverse factor appearing in an uncorrected measurement. They must not be interchanged when interpreting the common component of a later estimator.

## 1.2 Reference recovery, LO scaling and output synthesis

A DDS with $P$ phase bits and integer increment $I_L$ has nominal and physical LO frequencies

$$L = \frac{I_L}{2^P} F_0, \qquad L_{\text{act}} = \frac{I_L}{2^P} F = L \frac{F}{F_0}. \tag{10}$$

For the selected positive RF component, the nominal-scale downconverted frequency is

$$b_{\text{meas}} = F_{\text{DS},0}\, D \left( \frac{f}{F} - \frac{I_L}{2^P} \right) \tag{11}$$

$$= f \frac{F_0}{F} - L. \tag{12}$$

Consequently the complete nominal-scale reconstruction and its correction are

$$M = L + b_{\text{meas}} = f \frac{F_0}{F}, \qquad \boxed{f = \frac{F}{F_0}(L + b_{\text{meas}}).} \tag{13}$$

For a modal drive, the baseband term contains the configured center tone and its fitted line offset. Real excitation also has an LO-minus-baseband component; its physical axis uses the corresponding minus sign. Correcting only baseband while leaving the nominal LO unchanged would not recover physical RF.

The known reference supplies the clock estimate

$$M_R = L_R + b_R = f_R \frac{F_0}{F}, \qquad \boxed{\widehat{F} = F_0 \frac{f_R}{L_R + b_R}}, \qquad \widehat{C}_F = \frac{\widehat{F}}{F_0}. \tag{14}$$

In the comparison tables, $\widehat{F}_R$ denotes this reference-derived estimate of the board clock; $f_R$ denotes the known physical 10 MHz input. The original corrected modal vector was therefore

$$\mathbf{M}_m = \begin{bmatrix} M_{\text{C300}} \\ M_{\text{A100}} \\ M_{\text{C100}} \end{bmatrix}, \qquad \boxed{\mathbf{M}_{\text{corr}} = \widehat{C}_F \mathbf{M}_m.} \tag{15}$$

Using $\widehat{F}/D$ instead of $F_0/D$ on a baseband frequency axis is equivalent for that baseband term; full RF reconstruction must still include the physically scaled LO.

For a numerical example, let $L_R = 9\,999\,000$ Hz and $b_R = 717$ Hz:

$$M_R = 9\,999\,717 \text{ Hz}, \tag{16}$$

$$\widehat{F} = 65\,000\,000 \frac{10\,000\,000}{9\,999\,717} \simeq 65\,001\,839.6 \text{ Hz}, \tag{17}$$

$$\widehat{C}_F \simeq 1.0000283015. \tag{18}$$

The same estimate already provides reference-assisted clock generation:

$$f_{\text{out}} = \frac{I_{\text{NCO}}}{2^{26}} F, \qquad \boxed{I_R = \text{round}\left( 2^{26} \frac{f_*}{\widehat{F}} \right).} \tag{19}$$

Here $f_*$ is the desired physical output frequency. Equivalently, a nominal-domain frequency command is multiplied by $F_0/\widehat{F}$. This is the inverse of the measurement correction. The initial reference-driven NCO calculation did not require the temperature estimate as an input; autonomous synthesis was the later objective.

## 1.3 The initial one-state WLS estimator

Let $T_0 = 25\,^\circ\mathrm{C}$, $d = T - T_0$ and $f_{i,0} = f_i(T_0)$. The local characterization model is

$$f_i(T) = f_{i,0}(1 + a_i d), \qquad a_i = 10^{-6}\alpha_i. \tag{20}$$

The original input to the estimator explicitly included the reference scale factor:

$$p_i = 10^6 \left( \frac{M_{\mathrm{corr},i}}{f_{i,0}} - 1 \right) \tag{21}$$

$$= 10^6 \left( \frac{\widehat{F}}{F_0} \frac{M_{m,i}}{f_{i,0}} - 1 \right) = K_i \left( \frac{\widehat{F}}{F_0} M_{m,i} - f_{i,0} \right), \quad K_i = \frac{10^6}{f_{i,0}}. \tag{22}$$

The $K_i$ constants allow division-free ppm conversion at run time. The statistical model and cost are

$$\mathbf{p} = \boldsymbol{\alpha} d + \mathbf{e}, \qquad \mathbf{e} \sim \mathcal{N}(\mathbf{0}, \Sigma), \qquad Q = \Sigma^{-1}, \tag{23}$$

$$J(d) = (\mathbf{p} - \boldsymbol{\alpha} d)^T Q (\mathbf{p} - \boldsymbol{\alpha} d). \tag{24}$$

For independent errors, $\Sigma = \mathrm{diag}(\sigma_1^2, \sigma_2^2, \sigma_3^2)$ and

$$J(d) = \sum_{i=1}^{3} \frac{(p_i - \alpha_i d)^2}{\sigma_i^2}. \tag{25}$$

Differentiating and solving the normal equation gives

$$\frac{\partial J}{\partial d} = -2\boldsymbol{\alpha}^T Q (\mathbf{p} - \boldsymbol{\alpha} d) = 0, \tag{26}$$

$$\boxed{\widehat{d}_1 = (\boldsymbol{\alpha}^T Q \boldsymbol{\alpha})^{-1} \boldsymbol{\alpha}^T Q \mathbf{p}} = \frac{\sum_i \alpha_i p_i / \sigma_i^2}{\sum_i \alpha_i^2 / \sigma_i^2}, \tag{27}$$

$$W_1 = (\boldsymbol{\alpha}^T Q \boldsymbol{\alpha})^{-1} \boldsymbol{\alpha}^T Q, \quad \widehat{T}_1 = T_0 + W_1 \mathbf{p}, \quad W_1 \boldsymbol{\alpha} = 1. \tag{28}$$

For a correct, known residual covariance,

$$\mathrm{Var}(\widehat{d}_1) = (\boldsymbol{\alpha}^T Q \boldsymbol{\alpha})^{-1}. \tag{29}$$

This is a model uncertainty, not the observed spread of a temperature sweep. The first implementation assumed $\sigma_i = 1$ ppm, i.e. equal weights, pending covariance characterization.

## 1.4 The sawtooth in temperature: why decoupling was needed

With ideal simultaneous measurements, reference correction is exactly appropriate:

$$M_{m,i} = f_i(T)\frac{F_0}{F}, \qquad \widehat{F} = F \implies M_{\mathrm{corr},i} = f_i(T). \tag{30}$$

It should remove the clock fluctuation. The observed sawtooth temperature therefore indicated failure of this cancellation in the actual measurement/control chain, not a mathematical need to remove a valid correction arbitrarily.

Let $k = \widehat{C}_F s$ denote the remaining common multiplicative scale. Then

$$\frac{M_{\mathrm{corr},i}}{f_{i,0}} = k(1 + a_i d), \tag{31}$$

$$p_i = 10^6(k-1) + k\alpha_i d. \tag{32}$$

For $k = 1 + \delta$ and $c = 10^6\delta$, this is

$$\mathbf{p} = \boldsymbol{\alpha} d + \mathbf{1}c + \delta\boldsymbol{\alpha} d + \mathbf{e}. \tag{33}$$

Applying the one-state temperature row gives

$$\boxed{\widehat{d}_1 = (1+\delta)d + (W_1\mathbf{1})c + W_1\mathbf{e}.} \tag{34}$$

The dominant common-ppm leakage is $(W_1\mathbf{1})c$; the smaller multiplicative temperature term $\delta d$ remains explicit here. The condition $W_1\boldsymbol{\alpha} = 1$ preserves temperature sensitivity; it does not impose $W_1\mathbf{1} = 0$. The initial equal-weight model gives $W_1\mathbf{1} \simeq -0.03018\,^\circ\mathrm{C/ppm}$. In the later weighted configuration,

$$W_1 = (-0.327179055, -0.025307908, -0.011944104), \qquad W_1\mathbf{1} = -0.364431067\,^\circ\mathrm{C/ppm}. \tag{35}$$

Thus 10 ppm of an uncanceled common shift can appear as approximately $-3.64\,^\circ\mathrm{C}$ in the later WLS. This coefficient is calibration-dependent.

An early measurement sequence contains a transition with the following values:

| Quantity | Before transition | After transition |
|---|---|---|
| Estimated board clock (Hz) | 64,999,390 | 65,001,943 |
| Corrected C300 (Hz) | 10,003,855.900 | 10,004,246.666 |
| Corrected A100 (Hz) | 6,269,129.517 | 6,269,374.838 |
| Corrected C100 (Hz) | 3,388,383.686 | 3,388,515.623 |
| Logged $d = T - 25$ (°C) | 3.067 | 1.886 |

The corrected mode jumps are approximately $(39.062, 39.132, 38.938)$ ppm, close to the reference scale jump of 39.277 ppm. Removing the applied scale leaves raw modal changes of only a few hertz. A shared fractional shift is being mapped into temperature. The historical $d$ values use the earlier estimator; the later $-0.36443$ coefficient is not applied retrospectively to them.

**Design response.** Add a common-scale state or cancel scale through ratios, while continuing to investigate why the modal observations do not exhibit the expected inverse clock response. Decoupling temperature and recovering physical clock scale are related but separately testable objectives.

# 2. Joint and alternative estimators

## 2.1 Corrected joint WLS: temperature plus a common component

The first improvement retains corrected ppm observations, but adds a nuisance state:

$$\mathbf{p} = H\boldsymbol{\theta} + \mathbf{e}, \qquad H = \begin{bmatrix} \alpha_1 & 1 \\ \alpha_2 & 1 \\ \alpha_3 & 1 \end{bmatrix}, \qquad \boldsymbol{\theta} = \begin{bmatrix} d \\ c \end{bmatrix}. \tag{36}$$

The weighted cost and normal equations are

$$J(\boldsymbol{\theta}) = (\mathbf{p} - H\boldsymbol{\theta})^T Q(\mathbf{p} - H\boldsymbol{\theta}), \tag{37}$$

$$\nabla J = -2H^T Q(\mathbf{p} - H\boldsymbol{\theta}) = 0, \tag{38}$$

$$\boxed{\widehat{\boldsymbol{\theta}} = (H^T Q H)^{-1} H^T Q \mathbf{p}.} \tag{39}$$

For diagonal weights $w_i = 1/\sigma_i^2$, define the weighted sums

$$A = \sum_i w_i \alpha_i^2, \quad B = \sum_i w_i \alpha_i, \quad C = \sum_i w_i, \quad D_p = \sum_i w_i \alpha_i p_i, \quad E_p = \sum_i w_i p_i. \tag{40}$$

Then

$$H^T Q H = \begin{bmatrix} A & B \\ B & C \end{bmatrix}, \quad \Delta = AC - B^2 > 0, \quad \boxed{\widehat{d}_J = \frac{CD_p - BE_p}{\Delta}, \quad \widehat{c}_J = \frac{AE_p - BD_p}{\Delta}.} \tag{41}$$

Distinct temperature coefficients make the columns independent. The temperature row has elements

$$(W_T)_i = \frac{w_i(C\alpha_i - B)}{\Delta}, \qquad W_T\boldsymbol{\alpha} = 1, \qquad \boxed{W_T\mathbf{1} = 0}. \tag{42}$$

Consequently an additive common ppm shift changes $\widehat{c}_J$, while leaving $\widehat{d}_J$ unchanged. The fitted covariance under the assumed model is

$$\mathrm{Cov}(\widehat{\boldsymbol{\theta}}) = (H^T Q H)^{-1} = \frac{1}{\Delta} \begin{bmatrix} C & -B \\ -B & A \end{bmatrix}. \tag{43}$$

**Exact multiplicative interpretation.** If the actual observations obey $p_i = k\alpha_i d + 10^6(k-1)$, the ideal joint fit gives $\widehat{d}_J = kd$ and $\widehat{c}_J = 10^6(k-1)$. The logged corrected-joint temperature uses $25 + \widehat{d}_J$; first-order common-shift rejection is not exact multiplicative invariance. Under this ideal model one could instead recover

$$d = \frac{\widehat{d}_J}{1 + 10^{-6}\widehat{c}_J}. \tag{44}$$

The raw joint formulation below handles that ratio directly. Moreover, $\widehat{c}_J$ is a residual component *after reference correction*; it is not automatically the physical board-clock error and must not be applied again to the NCO. Evidence: `README.temperature-clock-compensation.md; corrected-joint fields in both retained runs.`

## 2.2 Raw joint WLS: solve temperature and scale directly

To avoid injecting the reference scale into temperature estimation, normalize the uncorrected modal observations by their characterization anchors:

$$z_i = \frac{M_{m,i}}{f_{i,0}} = s(1 + a_i d) + \eta_i, \qquad s = \frac{F_0}{F}. \tag{45}$$

Introduce $u = s$ and $v = sd$. This gives a linear fit in two parameters:

$$\mathbf{z} = B_z\boldsymbol{\beta} + \boldsymbol{\eta}, \quad B_z = \begin{bmatrix} 1 & a_1 \\ 1 & a_2 \\ 1 & a_3 \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} u \\ v \end{bmatrix}, \quad Q_z = \Sigma_z^{-1}. \tag{46}$$

Minimization yields

$$J_z = (\mathbf{z} - B_z\boldsymbol{\beta})^T Q_z (\mathbf{z} - B_z\boldsymbol{\beta}), \tag{47}$$

$$\frac{\partial J_z}{\partial \boldsymbol{\beta}} = -2B_z^T Q_z(\mathbf{z} - B_z\boldsymbol{\beta}) = 0, \tag{48}$$

$$\boxed{\widehat{\boldsymbol{\beta}} = (B_z^T Q_z B_z)^{-1} B_z^T Q_z \mathbf{z}.} \tag{49}$$

For independent normalized errors, let

$$S_0 = \sum_i w_i, \quad S_1 = \sum_i w_i a_i, \quad S_2 = \sum_i w_i a_i^2, \quad Z_0 = \sum_i w_i z_i, \quad Z_1 = \sum_i w_i a_i z_i. \tag{50}$$

The explicit solution and reconstructed states are

$$\Delta_z = S_0 S_2 - S_1^2, \tag{51}$$

$$\widehat{u} = \frac{S_2 Z_0 - S_1 Z_1}{\Delta_z}, \qquad \widehat{v} = \frac{S_0 Z_1 - S_1 Z_0}{\Delta_z}, \tag{52}$$

$$\boxed{\widehat{T}_{\text{raw}} = 25 + \frac{\widehat{v}}{\widehat{u}}}, \qquad \boxed{\widehat{F}_{\text{crystal}} = \frac{F_0}{\widehat{u}}}. \tag{53}$$

For fixed weights, multiplying every observation by $q$ multiplies both fitted parameters by $q$. Hence

$$\widehat{u}' = q\widehat{u}, \quad \widehat{v}' = q\widehat{v}, \qquad \widehat{v}'/\widehat{u}' = \widehat{v}/\widehat{u}. \tag{54}$$

Temperature is exactly invariant to an ideal common multiplicative scale within the stated linear crystal model. Clock scale is retained in $\widehat{u}$, rather than discarded. Model uncertainty propagates through

$$C_\beta = (B_z^T Q_z B_z)^{-1}, \quad g_T = \begin{bmatrix} -\widehat{v}/\widehat{u}^2 & 1/\widehat{u} \end{bmatrix}, \quad \operatorname{Var}(\widehat{T}) \simeq g_T C_\beta g_T^T. \tag{55}$$

No external-reference measurement enters this fit. Its uncertainty still depends on valid resonance observations and calibration; invariance does not establish absolute accuracy. Firmware can use ppm-scaled columns and precomputed coefficients rather than a poorly scaled run-time matrix inversion.

The candidate output word follows without an additional temperature factor:

$$\boxed{I_{\text{crystal}} = \operatorname{round}\left(\frac{2^{26} f_*}{F_0}\widehat{u}\right).} \tag{56}$$

This candidate was logged for comparison; it was not promoted to autonomous output control.

## 2.3 Ratio estimation and relative clock reconstruction

### Temperature from two modal frequencies

For modes $i$ and $j$, form the ratio of normalized raw measurements:

$$R_{ij} = \frac{M_{m,i}/f_{i,0}}{M_{m,j}/f_{j,0}} = \frac{s(1+a_i d)}{s(1+a_j d)} = \frac{1+a_i d}{1+a_j d}. \tag{57}$$

Rearranging gives

$$R_{ij}(1+a_j d) = 1 + a_i d, \tag{58}$$

$$R_{ij} - 1 = (a_i - R_{ij} a_j) d, \tag{59}$$

$$\boxed{\widehat{d}_{ij} = \frac{R_{ij}-1}{a_i - R_{ij} a_j}, \qquad \widehat{T}_{ij} = 25 + \widehat{d}_{ij}.} \tag{60}$$

The implementation uses A100/C300 because their temperature coefficients differ strongly. It cancels an ideal common scale exactly, but channel-specific errors, response offsets and calibration mismatch remain. Several ratios sharing the same mode have correlated errors; they are not independent extra measurements.

If $\sigma_R^2$ is known, first-order propagation gives

$$\frac{\partial d}{\partial R} = \frac{a_i - a_j}{(a_i - R a_j)^2}, \qquad \mathrm{Var}(\widehat{d}) \simeq \left[\frac{a_i - a_j}{(a_i - R a_j)^2}\right]^2 \sigma_R^2. \tag{61}$$

From a predicted temperature, each trusted mode could then independently recover the clock:

$$\widehat{F}_i = F_0 \frac{f_{i,0}(1 + a_i \widehat{d})}{M_{m,i}}, \qquad I_i = \mathrm{round}\left[\frac{2^{26} f_*}{F_0} \frac{M_{m,i}}{f_{i,0}(1 + a_i \widehat{d})}\right]. \tag{62}$$

### Relative joint fit used to test the clock waveform

The later reconstruction test avoids relying on absolute calibration offsets. With a trusted anchor capture at temperature $T_a$, define

$$y_i(t) = 10^6 \log \frac{M_{m,i}(t)}{M_{m,i}(t_a)}. \tag{63}$$

For small temperature changes and clock ratios,

$$y_i(t) \simeq \alpha_{i,\mathrm{eff}} \Delta T - \Delta\epsilon_F, \tag{64}$$

$$\alpha_{i,\mathrm{eff}} = \frac{\alpha_i}{1 + a_i(T_a - 25)}, \qquad \Delta\epsilon_F = 10^6 \log \frac{F(t)}{F(t_a)}. \tag{65}$$

Fit $H_{\mathrm{rel}} = [\boldsymbol{\alpha}_{\mathrm{eff}}, -\mathbf{1}]$ using the same normal-equation form as joint WLS. The retained relative comparisons use local equal-weight least squares, not reference-trained gains. The independent reference is an evaluation signal, not a fitted input. This relative diagnostic tests the missing waveform; it is distinct from the absolute raw joint temperatures below.

One can also model uncertain common correction statistically through $\Sigma_{\mathrm{tot}} = \Sigma_{\mathrm{modes}} + \sigma_c^2 \mathbf{1}\mathbf{1}^T$, if the common error is independent and zero mean. Such covariance inflation is a modeling option, not an additional estimator curve validated in these logs.

## 3. Recorded estimator comparison near 19 °C

The extended cooler run contains 1055 logged groups over 87.83 nominal minutes, with 674 complete groups available for all four temperature estimators. The context was a room reading near 19.5 °C; this is not an independently synchronized measurement of crystal temperature. All reported temperature offsets are converted to absolute temperature by adding 25 °C, except `temp_raw_joint`, which is already absolute.

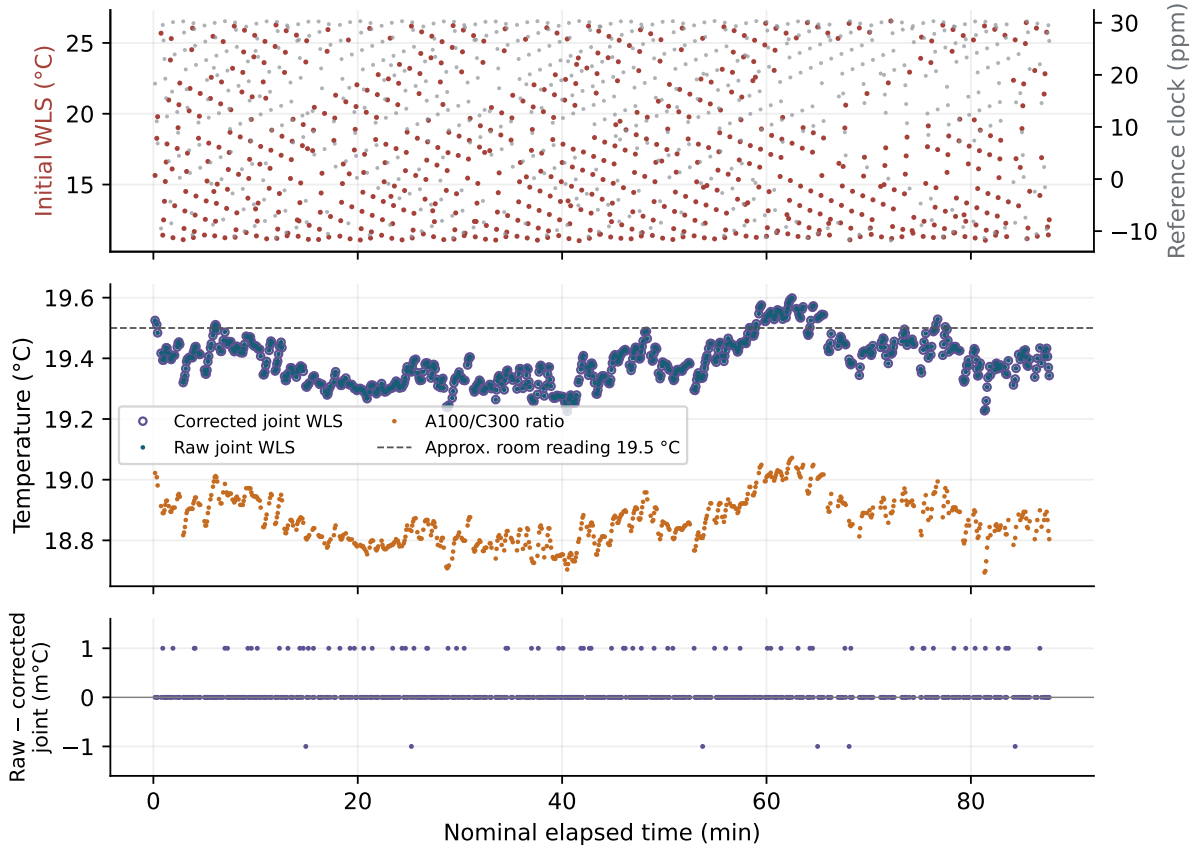



**Figure 1:** Cooler run. Top: initial one-state WLS temperature (red, left axis) and reference-derived board-clock offset (gray, right axis). Middle: corrected joint, raw joint and ratio temperatures. Bottom: raw-minus-corrected joint temperature at the precision of the saved log. Corrected and raw joint curves nearly coincide. Points are not interpolated across missing estimates.

| Estimator | Mean (°C) | Std (°C) | Correlation with $\widehat{F}_R$ |
|---|---|---|---|
| Initial WLS | 17.092 | 4.739 | −0.99990 |
| Corrected joint WLS | 19.387 | 0.0754 | +0.0760 |
| Raw joint WLS | 19.387 | 0.0754 | +0.0772 |
| A100/C300 ratio | 18.866 | 0.0761 | +0.0728 |

**Result.** Common-component rejection changes the interpretation of the same modal data: the initial temperature strongly follows the clock correction, while the joint and ratio estimates remain comparatively quiet. The raw joint estimate also removes the external reference from the temperature calculation. The joint-minus-ratio mean offset is 0.521 °C; their stability does not resolve that absolute calibration difference.

## 3.1 Recorded estimator comparison near 23 °C

The warmer comparison uses the 102 common timestamps, spanning 9.25 nominal minutes at the recorded five-second group cadence. It is a separate experiment, not a continuation of the cooler curve. The room reading was approximately 23 °C.

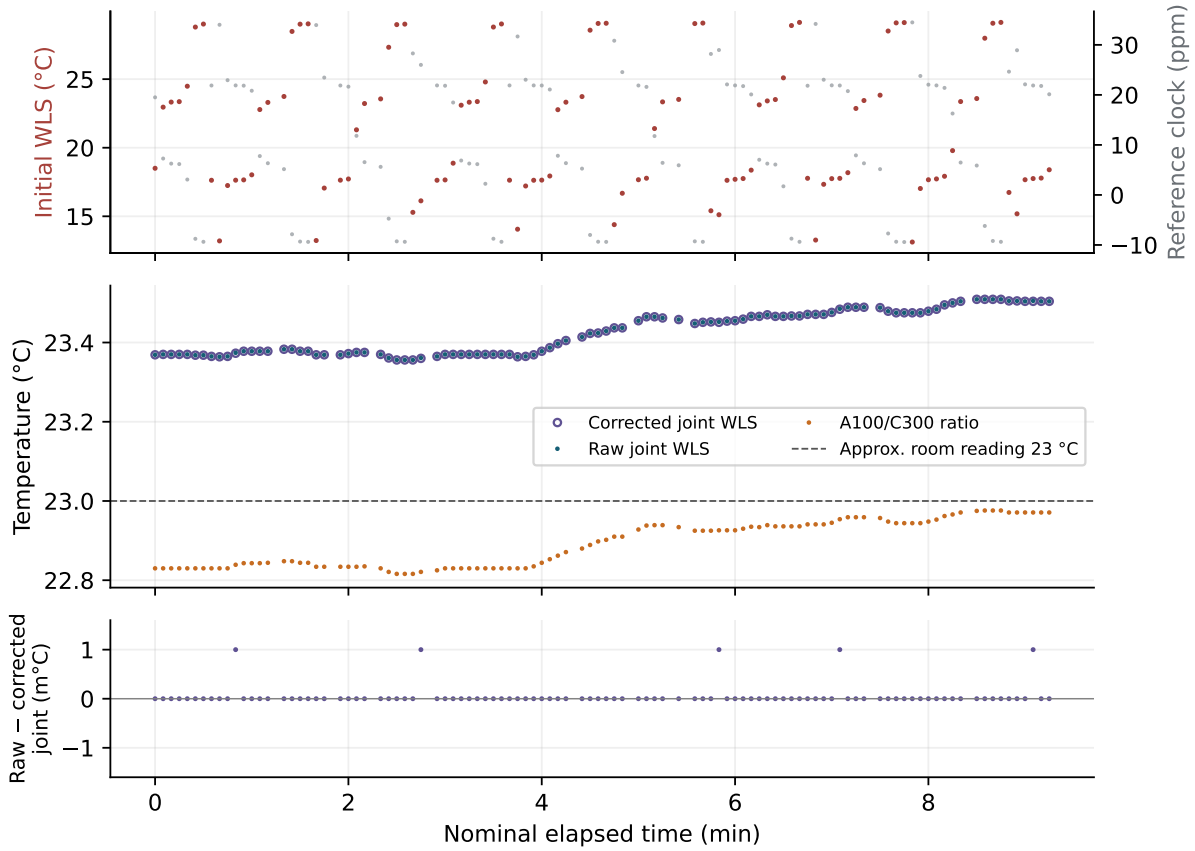



**Figure 2:** Warmer run, using the same four estimator definitions and panel arrangement. Initial WLS again follows the sawtooth correction. Corrected and raw joint temperatures overlap within the logged millidegree precision, while the ratio remains lower by approximately 0.53 °C.

| Estimator | Mean (°C) | Std (°C) | Correlation with $\widehat{F}_R$ |
|---|---|---|---|
| Initial WLS | 21.478 | 4.935 | −0.99995 |
| Corrected joint WLS | 23.425 | 0.0539 | +0.0373 |
| Raw joint WLS | 23.425 | 0.0539 | +0.0382 |
| A100/C300 ratio | 22.892 | 0.0571 | +0.0426 |

**Repeated outcome.** The decoupling effect persists at both approximate temperature levels. Raw joint temperature is warmer by about 4.04 °C between the two run means, while the joint/ratio separation remains near 0.52–0.53 °C. This supports the usefulness of the temperature-sensitive modal differences, without turning room agreement into an absolute calibration claim.

The corrected and raw joint estimates differ by no more than 0.001 °C in both retained comparisons at the saved logging precision. That agreement supports removal of the common scale from temperature; it does not independently validate the scale estimate itself. Both estimates share modal inputs, characterization assumptions and tracking limitations.

# 4. Tracking the modes: what is actually measured

Each mode is excited at three nearby frequencies. The controller compares left, center and right response powers and adjusts the LO toward a local response maximum. Selected spectral correlations and local interpolation are used; a full FFT is not required for modal tracking.

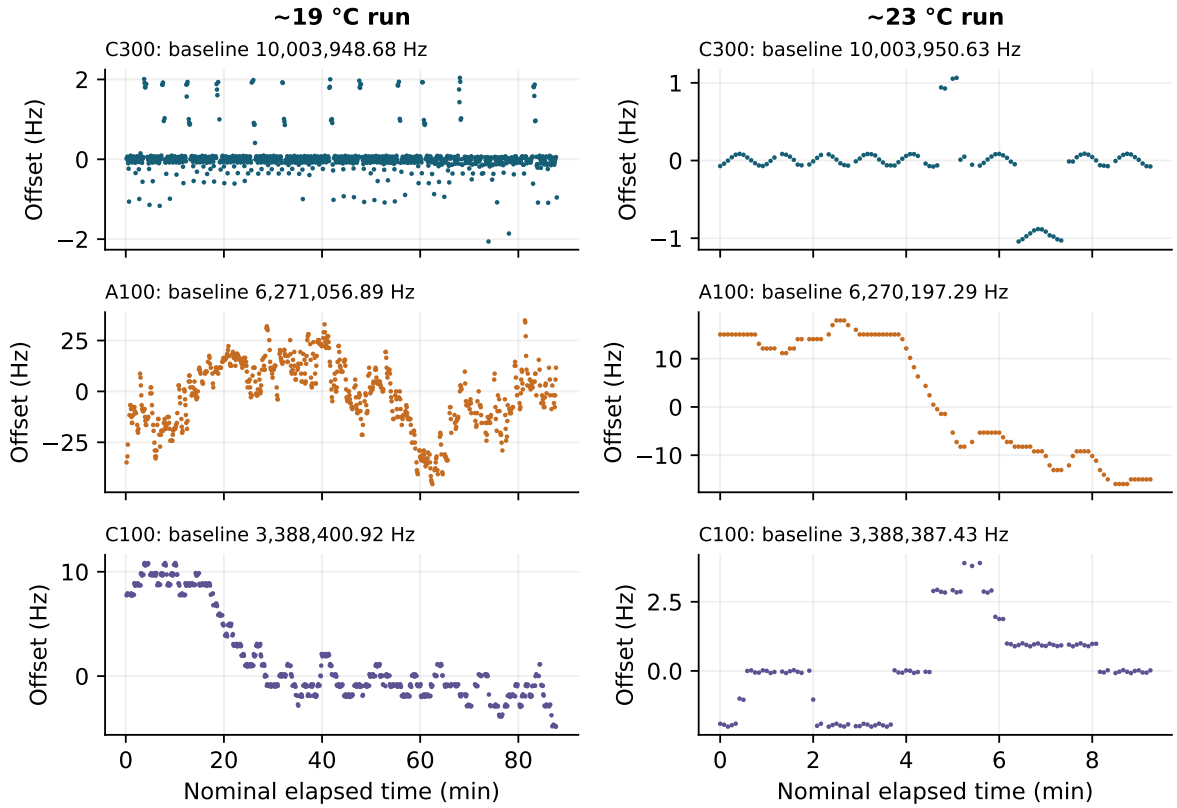



**Figure 3:** Tracked nominal-scale modal frequencies in the cooler run (left) and warmer run (right). Each panel subtracts its own stated baseline; the two experiments have different durations and vertical scales.

The logged modal frequency combines the LO active during capture, the configured baseband tone and the estimated central-line offset. It is expressed initially on the nominal 65 MHz board-clock scale. Its spectral estimate identifies the driven line.

| Mode | Approximate frequency | Temperature coefficient |
|---|---|---|
| C300 | 10.004 MHz | −0.359 ppm/°C |
| A100 | 6.270 MHz | −34.324 ppm/°C |
| C100 | 3.389 MHz | −1.172 ppm/°C |

The strong difference between A100 and the other slopes makes temperature distinguishable from a common fractional frequency shift. This is the useful foundation of the project.

**Measurement scope.** The cooler run covers approximately 87.8 nominal minutes with 674 complete three-mode groups; the warmer block has 102 common groups over 9.25 nominal minutes.

# 5. Establishing the board-clock sawtooth

Initially, the 10 MHz input on ADC2 was downconverted and estimated spectrally. An adapted Hann-window/log-power estimator reduced short-record bias; full-record spectral and zero-crossing regression estimates agreed to 0.0556 Hz RMS on steady portions. Both retained the sawtooth. This ruled out the original spectral interpolation as its sole cause, but both methods still shared the mixer/filter path.

To remove that dependence, a hardware counter was added before mixing and filtering. It counts UC clock cycles $C$ over $N$ periods of the raw external reference:

$$\boxed{\widehat{F}_s = C\,F_{\mathrm{ref}}/N}.$$

The result includes sequence and UC endpoint timestamps. This measurement does not rely on the nominal decimated sample rate or on locating a spectral bin.

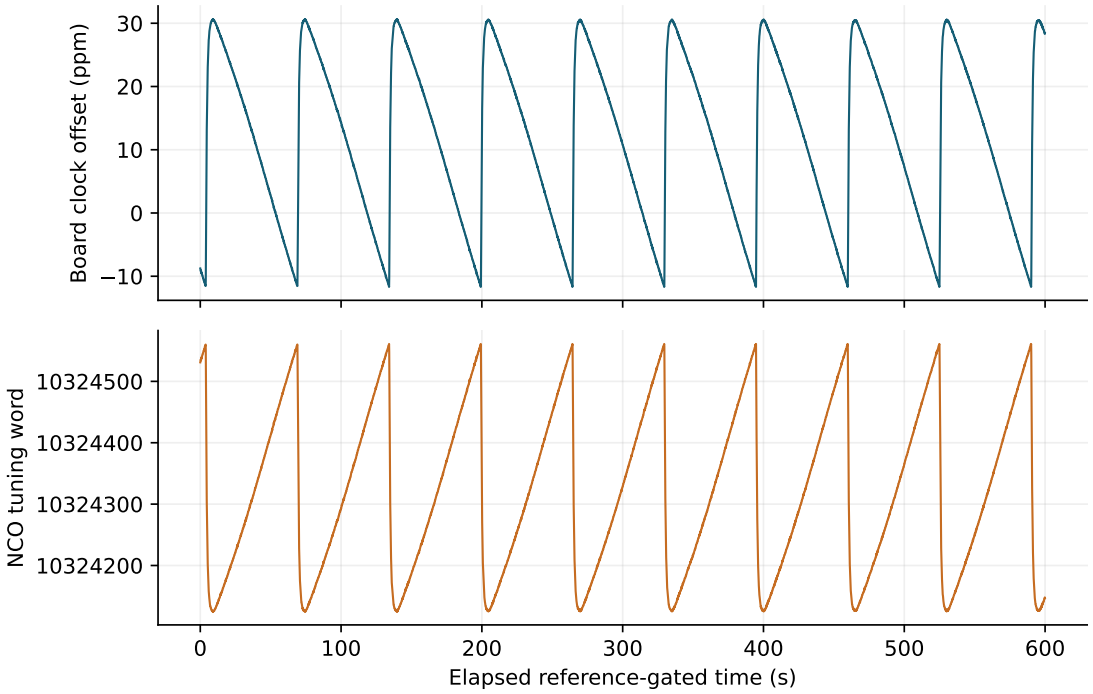



**Figure 4:** Raw hardware reference measurement and the inverse NCO correction during a ten-minute run. Each point represents a consecutive 100 ms reference gate. The tuning word is a control value, not a measurement of the DAC2 waveform.

| | |
|---|---|
| Consecutive accepted gates | 6000 |
| Missing / invalid / duplicate records | 0 / 0 / 0 |
| Measured UC frequency range | 64,999,240–65,001,990 Hz |
| Peak-to-peak relative change | 42.308 ppm |
| Observed recurrence | approximately 65.1–65.2 s |

**What this establishes.** Assuming the external 10 MHz input is stable, the sawtooth belongs to the board clock relative to that reference, rather than being an FFT or downconversion artifact. It does not identify the responsible component: the upstream oscillator, its compensation mechanism and the FPGA clock-generation chain have not been separated experimentally.

The reference-assisted controller updates $I_{\mathrm{NCO}}$ at about 10 Hz, independently of modal tracking. The same-gate computed residual is within approximately ±0.484 Hz at a 10 MHz target, consistent with tuning-word rounding. It excludes measurement/application delay and is **not evidence of measured output accuracy, phase error or jitter**. Independent physical DAC2 validation remains necessary.

# 6. Why autonomous clock reconstruction did not succeed

If the raw modal measurements followed physical resonances, a board-clock increase would produce an opposite fractional change in their nominal-scale values. The joint estimator should separate this common shift from temperature. The observed common shift is far too small and does not follow the independently measured sawtooth.

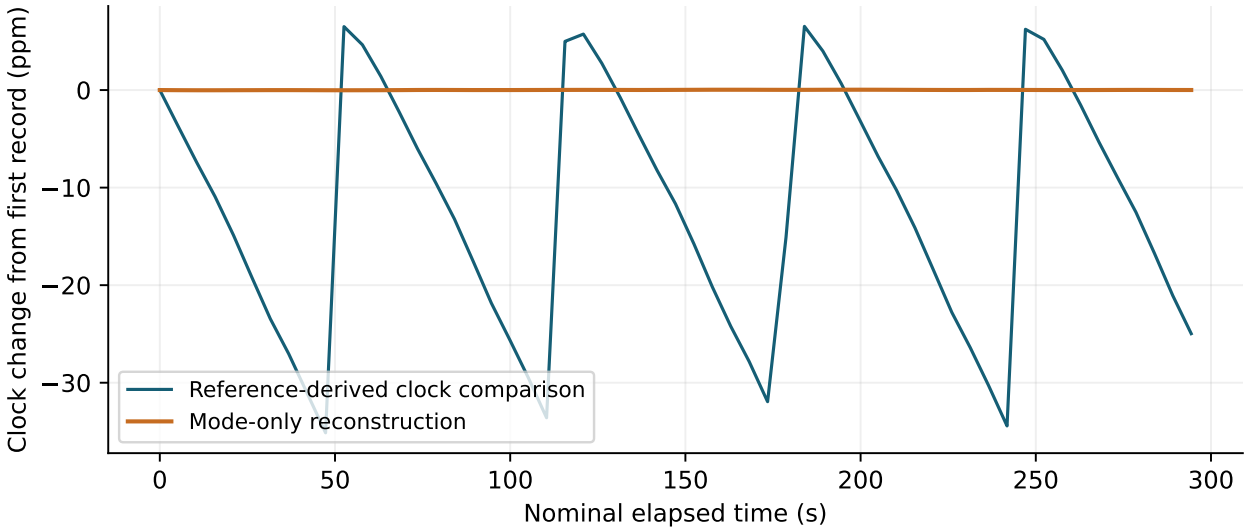



**Figure 5:** Final fresh tracking comparison: the reference-only clock change and the reference-free modal reconstruction, each relative to its first record. The reference is used for evaluation, not as an input to the modal fit.

| Comparison | Modal clock span | Reference clock span |
|---|---|---|
| Extended joint reconstruction | 0.356 ppm | 42.060 ppm |
| Final fresh tracking run | 0.0571 ppm | 41.666 ppm |

In the final run, correlation is $-0.0068$. Acquisition and excitation-underflow problems were corrected before this comparison. Adding the local three-tone detuning estimate did not recover the missing clock waveform. The failure therefore cannot be attributed solely to stale FIFO data, logging interruptions or the initial C100 position.

**Supported explanation: the measurements do not yet satisfy the reconstruction model.** TX and RX share the changing board clock. A driven spectral line can remain almost stationary in internal frequency units even while its physical RF frequency changes. The required information must then come from the crystal response and a sufficiently accurate resonance discriminator. A good spectral-line estimate alone cannot supply it.

**Remaining hypotheses.** Shallow or asymmetric response peaks, finite tracking bandwidth, amplitude/phase changes in the crystal path, and timing between reference estimation and excitation adjustment can bias the inferred resonance. Single-tone and fixed-inferred-RF experiments exposed response inconsistencies; strong and weak direct loopback controls were substantially more stable. These results narrow the problem but do not uniquely distinguish a moving resonance, analog-path dynamics and an incomplete RF/response model.

**Research conclusion.** The experiments demonstrate useful multimode thermometry and reference-assisted frequency correction. They do not demonstrate autonomous clock stabilization from the tracked modes. The negative result concerns the present measurement/control chain; it does not disprove the identifiability of temperature and clock scale from independent, correctly measured modal resonances. Retuning WLS weights cannot substitute for recovering that missing physical information.

# 7. Future direction and completion criteria

The next phase should connect measured phase dynamics, oscillator behavior and modal temperature information. It should begin with the external reference retained as an observation standard and proceed to reference-free tests only after each component is validated.

## A. Close and validate the phase relationship

Measure the generated DAC2 signal against the external reference with a coherent phase detector or suitable independent acquisition path. Record phase/time error, frequency error and cycle slips through several sawtooth resets. Compare a frozen tuning word, present frequency correction and a phase-error controller.

The current method corrects gate-average frequency while preserving accumulator continuity. Any residual frequency error integrates into phase drift. A phase loop would use measured phase error to steer the NCO; frequency matching alone cannot establish phase lock. Detector delay, loop bandwidth, gate averaging and NCO resolution must be characterized. Locking directly to the external reference is an intermediate validation step, not a reference-free solution.

**Success criterion:** bounded measured phase/time error through ramps and resets, with quantified frequency stability and no unexplained cycle slips. Compute output Allan deviation from an independent output measurement, not from the same gate used to choose its tuning word.

## B. Test whether sawtooth slope is predictable from temperature

Acquire long raw-counter records at controlled temperatures, including plateaus, warming/cooling trajectories and repeated power cycles. Use a sensor near the relevant oscillator during training: crystal temperature and FPGA-board oscillator temperature may differ or have different thermal delays.

Segment ramps and resets; estimate slope, recurrence, reset magnitude and their dependence on temperature and history. A candidate local predictor is

$$\widehat{F}_s(t + \Delta t) = \widehat{F}_s(t) + s(\widehat{T}, \text{history})\Delta t,$$

with a separately validated reset model. Temperature dependence of $s$ is a hypothesis, not a finding of the present work. Validate on held-out runs and compare against a temperature-independent predictor.

**Success criterion:** prediction reduces independently measured clock error over unseen ramps and resets. Slope alone is insufficient: an absolute frequency anchor, sawtooth state and reset timing are also needed. **Without a repeatable relationship between crystal-mode temperature and the board oscillator state, autonomous prediction may remain unobservable**.

# Final position

UberClock has established the thermometry concept, identified a real board/reference clock modulation and implemented a working reference-assisted correction path. The unresolved link is reliable observation of clock scale from the resonant modes. Phase-based validation and a tested temperature-dependent oscillator model are concrete next research directions; neither guarantees autonomous stabilization until the missing observability and physical output performance are demonstrated.